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தேர்வர் பெயர்

P **COMMON HALF YEARLY EXAMINATION - 2025**

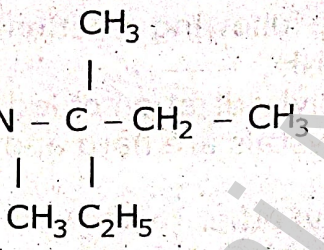
17.12.25 **Standard - 12** Reg.No. | | | | | | |

*Time Allowed:3.00 Hours* **CHEMISTRY** *Maximum Marks:70*

**I. Choose the correct answer:** **15×1=15**

1) The compound used as disinfectants are
a) Neomycin b) Erythromycin c) Chlorine compounds d) Cetirizine

2) Vitamin $B_2$ is also known as
a) Riboflavin b) Thiamine c) Nicotinamide d) Pyridoxine

3) IUPAC name for the amino


a) 3-Bimethylamino-3-methyl pentane
b) 3 (N,N-triethyl)-3-amino pentane
c) 3-N, N-Trimethyl pentan amine
d) N, N-dimethyl-3-methyl-pentan-3-amine

4) Weakest acid among the following is _____.
a) Acetic acid b) Phenol c) Water d) Acetylene

5) Assertion : Phenol is more acidic than ethanol.
Reason : Phenoxide ion is resonance stabilized.
a) Both A and R are true R is the correct explanation of A
b) A is true but R is false
c) Both A and R are false
d) None of these

6) Match the following:

| | | |
|---|---|---|
| a) $V_2O_5$ | - | i) High density polyethylene |
| b) Ziegler-Natta | - | ii) PAN |
| c) Peroxide | - | iii) $NH_3$ |
| d) Finely divided Fe | - | iv) $H_2SO_4$ |

**Code:**

| | A | B | C | D |
|---|---|---|---|---|
| a) | iv | i | ii | iii |
| b) | i | ii | iv | iii |
| c) | ii | iii | iv | i |
| d) | iii | iv | ii | i |

7) The electrolyte used in leclanche cell is
a) Paste of KOH and ZnO b) $NH_4Cl$ and $ZnCl_2$ in water
c) Li salt in organic solvent d) HgO mixed with Graphite

8) Which of the following is not a Lewis acid?
a) $SiCl_4$ b) $AlCl_3$ c) $SO_3$ d) CO

9) For a first order reaction, the rate constant is 6.909 $min^{-1}$, time taken for 75% conversion in minutes is
a) $\left(\frac{3}{2}\right)$ log 2 b) $\left(\frac{2}{3}\right)$ log 2 c) $\left(\frac{3}{2}\right)$ log $\left(\frac{3}{4}\right)$ d) $\left(\frac{2}{3}\right)$ log $\left(\frac{4}{3}\right)$

10) The ratio of close packed atoms to tetrahedral hole in cubic packing is
a) 1:1 b) 1:2 c) 2:1 d) 1:4

11) The co-ordination number of $Ni^{2+}$ in $Na_2$ [Ni (EDTA] complex is
a) 2 b) 4 c) 6 d) 8

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12) Which of the following is not coloured?
a) $Mn^{2+}$ b) $Zn^{2+}$ c) $Cr^{3+}$ d) $Cu^{2+}$

13) Formula for per nitrous acid
a) HOONO b) $H_2N_2O_2$ c) $HNO_2$ d) $HNO_4$

14) Oxidation state of carbon in its hydrides
a) +4 b) −4 c) +3 d) +2

15) Electrochemical process is used to extract
a) Iron b) Lead c) Sodium d) Silver

**II. Answer the following any six questions. Q. No.24 is compulsory: 6×2=12**

16) Which type of ores can be concentrated by froth floatation method? Give two examples for such ores.

17) Complete the following reactions.
i) $B(OH)_3 + NH_3 \longrightarrow$ ii) $B + NaOH \longrightarrow$

18) What is the hybridization of Iodine in $IF_7$? Give its structure.

19) Why ionic crystals are hard and brittle?

20) Define $P^H$.

21) How is diethyl ether reacts with following?
i) Low amount of HI ii) Excess of HI

22) Write a note on Wolf Kishner reduction.

23) Write the Esterification reaction.

24) The rate constant for a first order reaction is $1.54 \times 10^{-3} s^{-1}$, Calculate its half life time.

**III. Answer the following any six questions. Q.No.33 is compulsory: 6×3=18**

25) Write the chromyl chloride test.

26) Write the IUPAC name of the following.
i) $[Ag(NH_3)_2]Cl$ ii) $[FeF_6]^{4-}$ iii) $[Co(NH_3)_5Cl]^{2+}$

27) Explain why fluorine always exhibit an oxidation state of −1?

28) State Faraday's laws of electrolysis.

29) Write a note on catalytic poison.

30) Define Ionic product of water. Give its value at room temperature.

31) Give the difference between DNA and RNA.

32) How are tranquilizers works in body?

33) $C_6H_5NH_2 \xrightarrow[273K, HCl]{HNO_2} A \xrightarrow[Cu_2(CN)_2]{KCN} B \xrightarrow{H_3O^+} C$. Identify A, B, C.

**IV. Answer all the questions: 5×5=25**

34) a) Explain Forth flotation process. (5) **(OR)**
b) i) Write the preparation of potash alum. (3)
ii) Give the uses of Neon. (2)

35) a) Distinguish between Lanthanides and Actinides. (5) **(OR)**
b) Based on VB theory explain why $[Fe(CN)_6]^3$ is Paramagnetic, while $[Ni(CN)_4]^{2-}$ is diamagnetic. (5)

36) a) Calculate the percentage efficiency of packing in case of body centered cubic crystal. (5) **(OR)**
b) i) Derive Henderson Hessal Balch Equation. (3)
ii) Define order of the reaction. (2)

37) a) Explain adsorption theory of catalysis. (5) **(OR)**
b) i) Write the test to differentiate alcohols and phenols. (3)
ii) How will you prepare Malachitegreen from benzaldehyde? (2)

38) a) Explain the structure of glucose. (5) **(OR)**
b) i) Write a note on denaturation of proteins. (3)
ii) Write the preparation of Buna – N. (2)

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